



Modelling Financial Markets

Chapter 15: LLMs and Sentiment Analysis



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AMEX: American Stock Exchange
- API: Application Programming Interface
- AUC: Area Under the (ROC) Curve
- BERT: Bidirectional Encoder Representations from Transformers
- BH: Benjamini-Hochberg (procedure)
- BY-NC: Creative Commons Attribution-NonCommercial
- CI: Confidence Interval
- ECE: Expected Calibration Error
- EM: Expectation-Maximisation (algorithm)
- FDR: False Discovery Rate
- FNSPID: Financial News and Stock Price Integration Dataset
- FWER: Family-Wise Error Rate
- GI: (Harvard) General Inquirer, dictionary IV-4
- GPT: Generative Pre-trained Transformer
- GPT-1: Generative Pre-trained Transformer 1
- GPT-2: Generative Pre-trained Transformer 2
- GPT-3: Generative Pre-trained Transformer 3 (here version 3.5)
- GPT-4: Generative Pre-trained Transformer 4
- IV-4: Harvard IV-4 (version 4 of the General Inquirer word lists)
- LLM: Large Language Model
- LM: Loughran-McDonald (financial sentiment dictionary)
- LR: Logistic Regression
- MIT: Massachusetts Institute of Technology
- NASDAQ: National Association of Securities Dealers Automated Quotations
- NYSE: New York Stock Exchange

Acronyms used in this material (2/2)

- ▣ **OLS: Ordinary Least Squares**
- ▣ **PCA: Principal Component Analysis**
- ▣ **PPI: Prediction-Powered Inference**
- ▣ **RLHF: Reinforcement Learning from Human Feedback**
- ▣ **SE: Standard Error**
- ▣ **SESTM: Sentiment Extraction via Screening and Topic Modelling**
- ▣ **TF-IDF: Term Frequency - Inverse Document Frequency**
- ▣ **US: United States**
- ▣ **UTC: Coordinated Universal Time (Temps Universel Coordonné)**
- ▣ **VADER: Valence Aware Dictionary and sEntiment Reasoner**

Outline

Why Text?

Dictionaries

Labelled Data and Evaluation

Language Models: FinBERT and LLMs

Embeddings as Features

Look-Ahead Bias and Memorisation

From Sentiment to a Trading Signal

Sentiment Scores Are Estimates

Case Study: Can ChatGPT Forecast Stock Returns?

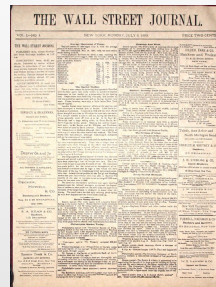
AI for Scientific Discovery

Practice and Conclusions

References

Markets Run on Words

- Prices move when information arrives, and most information arrives as text
 - ▶ News, company filings, earnings calls, analyst reports, central-bank statements, social media
- Financial newspapers are as old as organised markets
 - ▶ The Wall Street Journal: first issue on 8 July 1889 (photo)
- Today: millions of documents per day, far more than any analyst can read



The Wall Street Journal, first issue, 8 July 1889

Photo: Dow Jones & Company (1889); public domain;
[Wikimedia Commons](#)



Text as Data, and What We Mean by Sentiment

- Gentzkow, Kelly & Taddy (2019): a document d becomes numbers x_d (counts, scores, vectors), which feed an ordinary statistical model
 - ▶ Map x_d to a quantity of interest (sentiment, topic, risk), then use it in an economic question (returns, volatility)
- **Sentiment of a text**: good, bad or neutral news for the asset it mentions
 - ▶ Three classes, or the net score $s = P(\text{positive}) - P(\text{negative}) \in [-1, 1]$
 - ▶ Not **investor sentiment**, the mood of investors measured from market data (Baker & Wurgler (2006))
- Surveys for finance: Kearney & Liu (2014), Loughran & McDonald (2016)
- This chapter: the score is an **estimate** of the true tone, with measurement error; we then use it for daily stock returns

Early Evidence: Words Move Prices

- ▣ [Antweiler & Frank \(2004\)](#): internet message-board posts about stocks help predict volatility; the effect on returns is small
- ▣ [Tetlock \(2007\)](#): high media pessimism in a Wall Street Journal column predicts downward pressure on prices, followed by a reversal
 - ▶ Unusually high or low pessimism also predicts high trading volume
- ▣ [Tetlock, Saar-Tsechansky & Macskassy \(2008\)](#): the share of negative words in firm-specific news predicts lower earnings and returns; prices incorporate this information only gradually
- ▣ Common thread: text carries information that prices absorb, quickly but not always instantly

Narratives and Markets

- ▣ Shiller (2017): popular stories spread like epidemics and move economic decisions
 - ▶ Narratives can be measured only through text: newspapers, books, social media
- ▣ Shiller: Nobel Prize in Economics 2013, with Fama and Hansen
 - ▶ Chapter 2 (efficiency) and Chapter 17 (bubbles) meet here: narratives can push prices away from fundamentals



Robert J. Shiller, Stockholm, December 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; [Wikimedia Commons](#)

Four Ways to Turn Text into Sentiment

Method	How it works	Labels needed	Examples
Dictionary	count listed words	none	GI, LM, VADER
Fine-tuned model	network trained on labelled sentences	thousands	FinBERT
Large language model	instruction in plain language (prompt)	none (zero-shot)	Qwen2.5, GPT
Embeddings + classifier	text vectors into a regression	hundreds to thousands	MiniLM

- We compare all four on the same labelled data, with confidence intervals
- Then we ask whether any of them makes money on daily stock returns

Section Summary

- ▣ Most financial information is text; text becomes data through a numerical representation
- ▣ Text sentiment: the tone of a document about an asset, as classes or as a score in $[-1, 1]$
- ▣ Four families of methods, from word lists to large language models

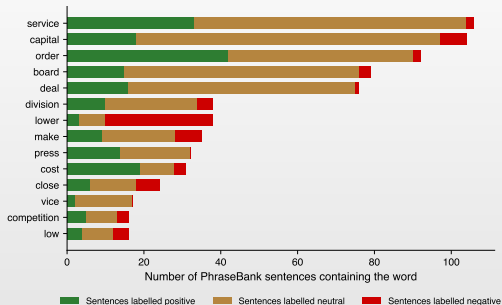
Bag of Words and Tone

- **Bag of words:** a document is reduced to its word counts; order is ignored
 - ▶ Tokens: words in capital letters, without digits and punctuation
- A dictionary lists positive words $\mathcal{P}$ and negative words $\mathcal{N}$
 - ▶ P_d, N_d : number of positive and negative words in document d
 - ▶ **Tone:** $\tau_d = \frac{P_d - N_d}{P_d + N_d} \in [-1, 1]$, and $\tau_d = 0$ if $P_d + N_d = 0$
 - ▶ **Class:** positive if $\tau_d > 0$, negative if $\tau_d < 0$, neutral otherwise
- Transparent, fast and free of training data; blind to context and negation

Generic versus Financial Dictionaries

- **Harvard General Inquirer (GI)**, list IV-4: a psychology dictionary from the 1960s
 - ▶ 2 007 negative and 1 637 positive words
- **Loughran & McDonald (2011) (LM)**: built from annual reports (10-K filings) of US firms
 - ▶ 2 355 negative and only 354 positive words; also lists for uncertainty and litigation
 - ▶ Their finding: almost three quarters of the GI negative words are not negative in a financial context
- In the lists used here: 1 552 GI negative words (77%) are not LM negative words

Negative for a Psychologist, Neutral for an Accountant



- Words negative in GI but not in LM, ranked by the number of Financial PhraseBank sentences that contain them
- Colour: the sentiment label that human annotators gave to the sentence

Interpretation: Finance Has Its Own Vocabulary

- "Service" (106 sentences), "capital" (104) and "board" (79) are business nouns, not bad news
 - ▶ Share of their sentences not labelled negative: 98%, 93%, 96%
 - ▶ "Vice" appears in "vice president": 100% of its sentences are not negative
- "Lower" is the exception: only 26% not negative, but "lower costs" is good news
 - ▶ The sign depends on the next word: a problem no word list can solve
- Of the 964 sentences whose only negative words are of this kind, 89% are not labelled negative

Worked Example: One Sentence, Two Dictionaries

- Financial PhraseBank sentence, labelled **positive** by all annotators:
 - ▶ *Uponor improved its performance in spite of the decrease in residential construction in the US .*
- LM: positive words improved; negative words –
 - ▶ $\tau = (1 - 0)/(1 + 0) = +1.00$: positive, correct
- GI: positive words –; negative words spite, decrease
 - ▶ $\tau = (0 - 2)/(0 + 2) = -1.00$: negative, wrong
 - ▶ GI misses "improved" and counts "spite" and "decrease", which describe the market, not the firm

VADER: Rules on Top of a Word List

- [Hutto & Gilbert \(2014\)](#): a lexicon with intensity scores for about 7,500 terms, built for social media
 - ▶ Rules for negation ("not good"), intensifiers ("very"), capital letters, exclamation marks and emoticons
 - ▶ Output: a compound score in $[-1, 1]$; usual cut-offs ± 0.05 for positive and negative
- Rules fix some context problems, but the vocabulary is still generic, not financial
 - ▶ It is the second generic benchmark in our tests, next to GI

Section Summary

- Dictionary tone: $(P - N)/(P + N)$ from counts of listed words
- Generic dictionaries misread finance: most GI negative words are ordinary business words
- LM fixes the vocabulary, not the context: "lower costs" still counts as negative

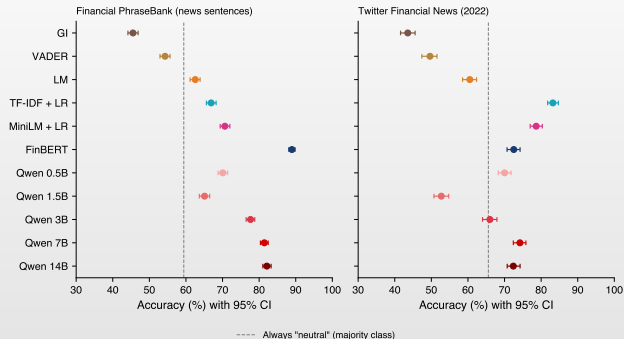
Two Labelled Corpora

- Financial PhraseBank ([Malo et al. \(2014\)](#)): 4 846 news sentences labelled by a pool of 16 annotators with a finance background, 5 to 8 per sentence
 - ▶ Question: good, bad or neutral news for the investor? Agreement levels: at least 50%, 66%, 75% or 100% of the sentence's annotators
 - ▶ 2 264 sentences with full agreement, 629 with agreement from 50% to below 66%; labels 59% neutral, 28% positive, 12% negative
- Twitter Financial News (MIT licence, Hugging Face Hub): headlines from 2022, labelled bearish, bullish or neutral
 - ▶ Training set of 9,543, validation set of 2 388 headlines (used for evaluation); labels 66% neutral, 20% positive, 15% negative
 - ▶ Short, noisy texts with tickers and links: closer to the real news flow
- Two data sets, because a result on one set may not carry over to another

Inference for Classifiers

- Accuracy, precision, recall and macro-F1 as in Chapter 13; benchmark: always "neutral"
 - ▶ It scores 66% accuracy on Twitter, but a macro-F1 of only 26.4%
- **McNemar test**: two classifiers on the same texts
 - ▶ n_{01} : A wrong and B right; n_{10} : A right and B wrong; texts where both agree carry no information
 - ▶ Under H_0 (equal accuracy), $n_{01} \sim \text{Binomial}(n_{01} + n_{10}, 0.5)$: an exact test
- Every test here treats the human label as the truth; with 5 to 8 annotators per sentence who often disagree, it is itself noisy
 - ▶ Label errors can cap measured accuracy; only under symmetric noise, independent of the classifiers, do they also shrink measured differences towards 0 (class-dependent noise can enlarge or reverse them)

Eleven Classifiers, Two Data Sets

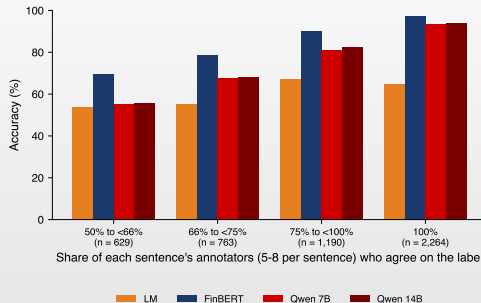


- TF-IDF + LR and MiniLM + LR: logistic regressions trained on the Twitter training set; all other methods use no Twitter labels

Interpretation: No Single Winner

- ▣ Generic dictionaries lose to the majority class on both sets: GI 43.6%, VADER 49.6% on Twitter
 - ▶ LM is better (60.5%) but still below "always neutral" (66%) on Twitter
- ▣ FinBERT: 89.0% on PhraseBank, 72.5% on Twitter; Qwen2.5-7B zero-shot: 81.4% and 74.2%
 - ▶ The best on Twitter is the simplest supervised model: TF-IDF + logistic regression, 83.2%
 - ▶ TF-IDF: counts of words and word pairs, weighted down when a term appears in many documents
- ▣ Lesson: labelled data from the target domain beat model size

When Humans Disagree, Models Disagree Too



- Financial PhraseBank split by the highest agreement threshold reached among each sentence's annotators; lowest group: 50% to below 66%

Interpretation: Ambiguity Sets a Ceiling

- ▣ Full agreement: FinBERT 97%, Qwen2.5-14B 94%, LM 65%
 - ▶ Agreement from 50% to below 66%: FinBERT 70%, Qwen2.5-14B 55%, LM 54%
- ▣ Many sentences are genuinely ambiguous: even experts split
 - ▶ A model cannot be judged on a label that humans do not agree on
- ▣ For signals: a strong, unambiguous tone is more informative than a borderline one

When Annotators Disagree: Agreement and Latent Labels

- **Fleiss' κ** (Fleiss (1971)): N texts, m raters, n_{ij} raters giving text i label j
 - ▶ $\bar{P} = \frac{1}{N} \sum_i \sum_j \frac{n_{ij}(n_{ij}-1)}{m(m-1)}$, $\bar{P}_e = \sum_j p_j^2$, $\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$; p_j : share of all votes for j
 - ▶ The formula assumes the same m for every text; PhraseBank has 5–8 raters per sentence and publishes only agreement levels, not the votes: κ cannot be computed from the release
- **Dawid–Skene model** (Dawid & Skene (1979)): rater a has a confusion matrix $\pi_{jl}^{(a)} = P(\text{says } l \mid \text{true } j)$; the true class is latent
 - ▶ EM estimates the $\pi^{(a)}$ and gives each text a soft label $P(\text{true} = j \mid \text{votes})$
- Evaluate models by a proper scoring rule (log loss) against soft labels, not by accuracy against the majority label
 - ▶ The ceiling of the previous slide is a label-noise ceiling, not a model failure

Section Summary

- ▣ Two labelled data sets: Financial PhraseBank (news sentences) and Twitter Financial News (headlines)
- ▣ Report accuracy with a CI and macro-F1; compare classifiers with the paired McNemar test
- ▣ Generic dictionaries fail; supervised models from the target domain are hard to beat

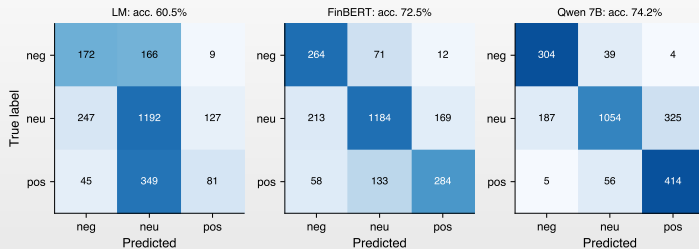
From Words to Words in Context

- A dictionary gives "lower" one fixed sign; a language model reads the sentence around it
 - ▶ Each token gets a vector that depends on all other tokens (attention, Chapter 14; [Vaswani et al. \(2017\)](#))
- **BERT** ([Devlin et al. \(2019\)](#)): a Transformer encoder pretrained on books and Wikipedia
 - ▶ Pretraining task: guess words hidden at random in a sentence (masked language modelling)
 - ▶ About 110 million parameters in the base version
- Fine-tuning: add a small classification layer and train the whole network on labelled sentences

FinBERT

- ▣ [Araci \(2019\)](#): BERT further pretrained on financial news, then fine-tuned for sentiment
 - ▶ The public model used here (ProsusAI/finbert) was fine-tuned on Financial PhraseBank
 - ▶ Output: probabilities of positive, negative and neutral
- ▣ [Huang, Wang & Yang \(2023\)](#): a FinBERT trained on analyst reports, earnings-call transcripts and 10-K filings
 - ▶ Outperforms dictionaries and other machine-learning methods, mainly on sentences that dictionaries misread
- ▣ Runs on an ordinary laptop: a few milliseconds per sentence

Where the Errors Are



□ Twitter validation set: rows are the true labels, columns the predicted labels; counts in the cells

Interpretation: Different Models, Different Mistakes

- ▣ LM sends most positive headlines to neutral: its positive list is short (354 words)
- ▣ FinBERT errs in both directions around neutral
- ▣ Qwen2.5-7B finds almost all positive and negative headlines, but calls many neutral ones positive or negative
 - ▶ An LLM "sees" sentiment where annotators saw plain facts: high recall, low precision on the extreme classes
- ▣ For trading, false positives cost money: precision on the extreme classes matters

The In-Sample Trap

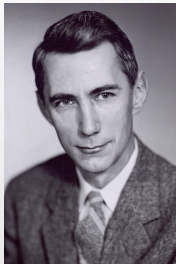
- ▣ FinBERT: 89.0% on PhraseBank, most of which it was trained on ([Araci \(2019\)](#): 3,101 of 4,845 sentences); 72.5% on Twitter: a drop of 16.4 points (differences from unrounded accuracies)
 - ▶ Qwen2.5-7B, not fine-tuned on these labels, also drops by 7.2 points: Twitter headlines are harder
- ▣ Difference in differences: $\Delta = (a_{F,P} - a_{F,T}) - (a_{Q,P} - a_{Q,T}) = 9.2$ points, bootstrap CI 95% [6.7; 11.9]
 - ▶ $a_{m,D}$: accuracy of model m (F: FinBERT, Q: Qwen) on data D (P: PhraseBank, T: Twitter)
 - ▶ Identifying assumption: without training overlap, both models would lose the same amount from PhraseBank to Twitter
 - ▶ Under this assumption, 9.2 of the 16.4 points come from training overlap; the CI measures sampling noise, not the assumption (the two architectures may react differently to a new domain)
- ▣ Rule: evaluate a language model only on texts it has never seen, as in the out-of-sample tests of Chapters 8 and 13

Worked Example: Is FinBERT Better than LM?

- ▣ Twitter validation set, 2 388 headlines: LM 60.5%, FinBERT 72.5%
 - ▶ LM wrong and FinBERT right: $n_{01} = 539$; LM right and FinBERT wrong: $n_{10} = 252$
 - ▶ Exact McNemar $p < 0.001$
- ▣ Difference +12.0 percentage points, bootstrap CI 95% [+10.0; +14.2]
 - ▶ Clear win for the fine-tuned model, even outside its training data

From Next-Word Prediction to Instructions

- **Shannon (1951)**: English is highly predictable from its past letters; a large language model (LLM) estimates $P(w_{t+1} \mid w_1, \dots, w_t)$ over tokens
 - ▶ **Brown et al. (2020)**: at scale, tasks are solved from a few examples in the prompt (few-shot)
 - ▶ **Ouyang et al. (2022)**: instruction tuning and RLHF teach the model to follow requests
- **BloombergGPT (Wu et al. (2023))**: 50 billion parameters, financial and general text; open general models now match it on sentiment
 - ▶ **Background**: [Quantinar: The Next Word Problem](#), [Quantinar: Natural Language Processing](#)



Claude E. Shannon, founder of information theory

Photo: [Tekniska museet, Stockholm \(undated\)](#); CC

BY 2.0; [Wikimedia Commons](#)

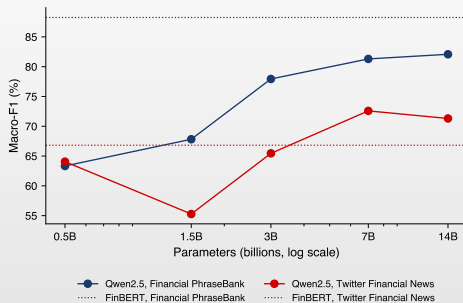
Zero-Shot Classification with an LLM

- Prompt: a system instruction plus the text to classify
 - ▶ You classify the sentiment of financial news for investors. Answer with one word: positive, negative or neutral.
 - ▶ Headline: <text> Sentiment:
- We read the probabilities of the three answer words at the first generated position
 - ▶ Logits z_k of the tokens "positive", "negative", "neutral": $p_k = e^{z_k} / \sum_j e^{z_j}$
 - ▶ Label: $\arg \max_k p_k$; score: $p_{\text{pos}} - p_{\text{neg}}$; no text is generated, so nothing needs parsing

The Models: Qwen2.5-Instruct

- ▣ [Qwen Team \(2024\)](#): open-weight models from 0.5 to 72 billion parameters, weights published on 19 September 2024
 - ▶ We use 0.5B, 1.5B, 3B, 7B and 14B (B: billion parameters), all with public weights; 3B is under the Qwen Research License, the others under Apache 2.0
- ▣ Wall-clock time per headline on a laptop graphics processor, including the prompt; median of one pass per prompt
 - ▶ 0.5B: 4 ms; 1.5B: 12 ms; 3B: 20 ms; 7B: 92 ms; 14B: 105 ms
 - ▶ Single passes are noisy (memory swapping, other load): 7B took 51 to 94 ms, 14B 88 to 112 ms; a timing claim needs repeated runs
 - ▶ Open weights: results can be reproduced, and the release date bounds what the model could know

Bigger Is Not Always Better



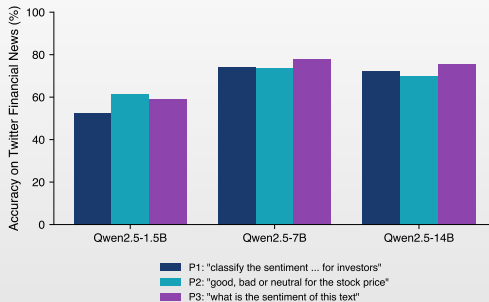
Macro-F1 of zero-shot Qwen2.5 with the same prompt; dotted lines: FinBERT

 MFM_ch15_llm_zero_shot

Interpretation: Size Helps, Unevenly

- ▣ PhraseBank: macro-F1 rises from 63.3% (0.5B) to 82.1% (14B), still below FinBERT (88.3%)
 - ▶ FinBERT's lead there is partly training overlap (The In-Sample Trap)
- ▣ Twitter: not monotone; 1.5B is the worst (55.3%), 7B the best (72.6%), above FinBERT (66.8%)
 - ▶ 7B against 1.5B on accuracy: +21.5 points, McNemar $p < 0.001$
- ▣ Small models react strongly to wording: the 1.5B model rarely answers "neutral" with this prompt

Same Model, Different Wording

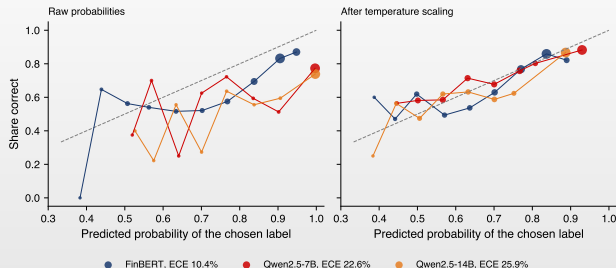


- Three prompts that ask the same question; Twitter validation set

Interpretation: the Prompt Is a Parameter

- ▣ Accuracy ranges: 1.5B from 52.7% to 61.7%; 7B from 73.7% to 77.8%; 14B from 69.9% to 75.6%
 - ▶ The three prompts give the same label on only 67% of headlines (7B)
 - ▶ Share of "neutral" answers for 7B: 48% (P1) to 69% (P3)
- ▣ Choosing the prompt after seeing test results is a form of data snooping (Chapter 13)
 - ▶ Fix the prompt in advance, or choose it on a separate validation set

Are the Probabilities Calibrated?



- Twitter validation set, one random half; bins of the probability of the chosen label; point size: number of headlines
- Right panel: $p_k(T) \propto p_k^{1/T}$ with T fitted by maximum likelihood on the other half (Guo et al. (2017))

 MFM_ch15_valid_inference

Interpretation: LLM Probabilities Are Overconfident

- ECE (expected calibration error): $\sum_b \frac{n_b}{n} |\text{acc}_b - \text{conf}_b|$ over 10 equal-width bins b of the top probability on $[1/3, 1]$
 - ▶ FinBERT 10.4%, Qwen2.5-7B 22.6%, Qwen2.5-14B 25.9%: 7B gives 88% of headlines a confidence above 0.93 (mean 99.7%) and is right on 77% of them
- Temperature scaling: $T = 8.4$ for 7B, 1.4 for FinBERT; ECE falls to 4.3% and 4.6%
 - ▶ Log loss $-\frac{1}{n} \sum_i \ln p_{iy_i}$ (probabilities floored at 10^{-12}); Brier $\frac{1}{n} \sum_i \sum_k (p_{ik} - 1\{y_i = k\})^2$, summed over the 3 classes
 - ▶ Log loss of 7B: 2.61 to 0.63; Brier score (Brier (1950)) 0.468 to 0.366; the labels, hence accuracy, do not change
- Calibration depends on the prompt: ECE of 7B is 22.6%, 22.1% and 19.8% under P1, P2, P3
 - ▶ The net score $p_{\text{pos}} - p_{\text{neg}}$ enters regressions: with raw LLM probabilities it is close to a three-valued label

Worked Example: LLM versus FinBERT

- ▣ Twitter, prompt P1: Qwen2.5-7B 74.2%, FinBERT 72.5%
 - ▶ $n_{01} = 395$, $n_{10} = 355$; McNemar $p = 0.154$; difference +1.7 points, CI 95% [-0.5; +3.8]
 - ▶ Qwen2.5-14B against FinBERT: -0.1 points, $p = 0.942$
- ▣ No significant difference detected, which is not proof of equal accuracy (the CI allows up to +3.8 points): 70 to 130 times more parameters and slower processing bring no demonstrated gain
 - ▶ Where LLMs help: no labels needed, any language, longer and more complex documents

Beyond Headlines: Long Documents

- ▣ Annual reports: [Loughran & McDonald \(2011\)](#) built their lists from 10-K filings; tone and uncertainty predict returns and volatility around filings
- ▣ Earnings calls and analyst reports: [Huang, Wang & Yang \(2023\)](#) show FinBERT extracts information that dictionaries miss
- ▣ [Kim, Muhn & Nikolaev \(2023\)](#): LLM summaries of disclosures are much shorter but explain stock-market reactions better than the originals
 - ▶ They measure disclosure "bloat": long, uninformative text is linked to lower price efficiency
- ▣ Long documents need models with long context windows, or chunking and aggregation

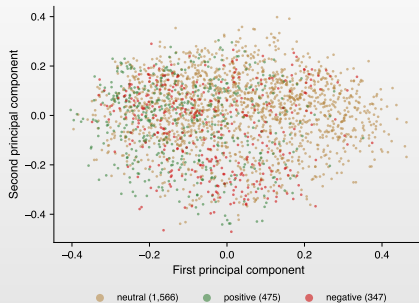
Section Summary

- ▣ FinBERT beats every dictionary out of sample; its PhraseBank score is inflated by training overlap (about 9.2 points under the identifying assumption)
- ▣ An LLM predicts the next token; zero-shot sentiment reads the probabilities of the answer words
- ▣ Accuracy grows with size on average, but not monotonically; the prompt changes the result
- ▣ On short financial headlines, a 7B to 14B open model shows no significant accuracy difference from FinBERT; its probabilities are overconfident until rescaled

Text Embeddings

- **Embedding:** a vector $e_d \in \mathbb{R}^k$, close for similar texts: word2vec (Mikolov et al. (2013)), Sentence-BERT (Reimers & Gurevych (2019))
 - ▶ Here MiniLM (Wang et al. (2020)), $k = 384$, unit length: similarity $\cos(e_a, e_b) = e_a^\top e_b$
- Embeddings enter an ordinary classifier or regression, like any other feature

What Embeddings Look Like

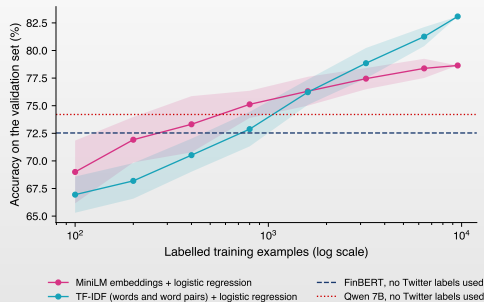


- Twitter validation headlines: first two principal components of the 384-dimensional MiniLM vectors, coloured by label

Interpretation: Sentiment Is Not the Main Direction

- The first two components separate topics (companies, macro news, crypto), not sentiment
 - ▶ PCA (Chapter 3) keeps the directions of largest variance, which need not be the ones we care about
- A supervised model finds the sentiment direction among the 384 dimensions
 - ▶ Logistic regression on all dimensions: 78.6% accuracy on Twitter

How Many Labels Do You Need?



- Logistic regression trained on random subsets of the Twitter training set, 10 repetitions per size; band: ± 2 standard deviations

Interpretation: a Few Hundred Labels Beat FinBERT

- ▣ Embeddings + logistic regression: 69.0% with 100 labels, above FinBERT from 400 labels
 - ▶ The pretrained vectors already contain most of what is needed
- ▣ TF-IDF starts lower (66.9%) but keeps improving: 83.1% with all 9,543 labels, against 78.6% for embeddings
 - ▶ With many labels, exact words ("upgrade", "cuts", "misses") carry the signal in headlines
- ▣ Practical rule: label a few hundred documents from your own domain before trusting any off-the-shelf model

Text Regressions with Statistical Theory

- [Taddy \(2013\)](#): multinomial inverse regression: model word counts given the outcome, $c_d \sim \text{MN}(q(y_d), m_d)$, c_d : counts, $m_d = \sum_j c_{dj}$: length
 - ▶ Log-linear model $q_j(y) \propto e^{\alpha_j + \varphi_j^\top y}$, loadings $\Phi = (\varphi_1, \dots, \varphi_p)^\top$: given m_d and this model, $z_d = \Phi^\top c_d / m_d$ is sufficient for y_d
- [Ke, Kelly & Xiu \(2019\)](#) (SESTM): screen words correlated with returns, fit a two-topic model of positive and negative words, score new articles by penalised likelihood
 - ▶ The paper gives statistical guarantees for the estimator; the signal is strongest on the day of and the day after the news
- [Bybee et al. \(2024\)](#): news attention to topics in 800,000 Wall Street Journal articles tracks economic activity and forecasts aggregate stock returns
 - ▶ Also [Jegadeesh & Wu \(2013\)](#) (word weights from 10-K reactions); rule for all: estimate on data before the forecast period

Section Summary

- ▣ Embeddings turn texts into vectors that enter ordinary regressions
- ▣ Sentiment is one direction among many; supervision finds it
- ▣ A few hundred in-domain labels beat a generic fine-tuned model; many labels favour simple word features

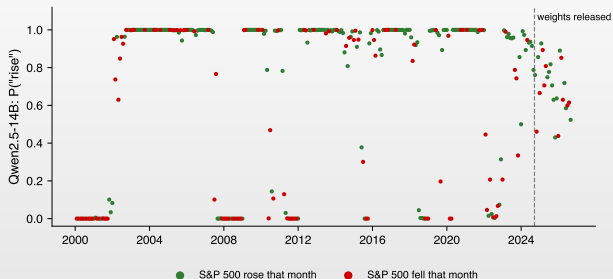
The Model May Know the Answer

- An LLM trained on text up to 2024 may have read what happened after a 2015 headline
 - ▶ A backtest on 2015 news then mixes forecasting with remembering
- Glasserman & Lin (2023): two biases in sentiment backtests
 - ▶ Look-ahead: the model knows the returns that followed the news
 - ▶ Distraction: general knowledge about the named company interferes with reading the text; in their tests it mattered more than look-ahead
- Sarkar & Vafa (2024): direct tests show look-ahead in LLM applications to earnings calls and elections

Our Test: Ask About the Past, Without Context

- Monthly: "Did the S&P 500 index rise or fall during <month year>?", every month from 296 pre-release months plus 23 after
 - ▶ Daily: "Did Apple stock close higher or lower on <date> than on the previous trading day?", 500 random days 2010–2023 and 493 days after October 2024
- Score: $P(\text{rise})$ from the answer-word probabilities; true direction from the saved price series
 - ▶ AUC: probability that a random up period gets a higher score than a random down period; 0.5 = no discrimination (Chapter 13)
- Outcomes after the release cannot be memorised in fixed weights: if the date itself carries no predictive information (e.g. seasonality), the population AUC is 0.5 there, while the sample AUC varies around it

What Qwen2.5-14B "Remembers" About the S&P 500

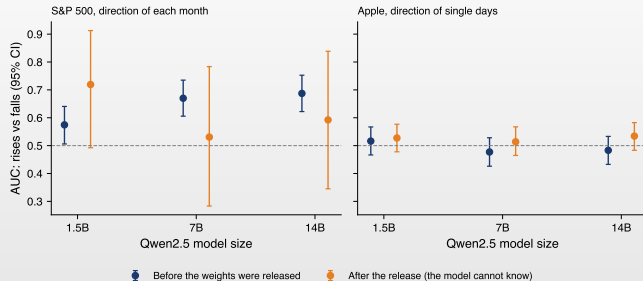


□ Probability of the answer "rise" for each month, no context given; colour: what the index actually did

Interpretation: It Remembers Regimes, and Some Famous Months

- "Fall" for 2000–2002 and 2008–2009, "rise" for most of 2003–2007 and 2013–2021: the bear and bull markets
 - ▶ October 2008 (-16.9%), March 2020 (-12.5%), December 2018 (-9.2%): $P(\text{rise}) \approx 0$
 - ▶ But September 2002 (-11.0%) and February 2009 (-11.0%) get $P(\text{rise})$ close to 1
- After the release the answers drift towards "rise": no knowledge, only the general belief that markets go up
- Knowledge is coarse: memorable episodes, not every month

Memory Measured: AUC Before and After Release



- Bootstrap 95% intervals, months resampled independently; 0.5 means the model cannot tell up from down

Interpretation: Months Yes, Days No

- ▣ Months before release: AUC 0.67 (7B, CI [0.61; 0.73]) and 0.69 (14B, CI [0.62; 0.75])
 - ▶ Permutation p -value for 14B: <0.001 ; the 1.5B model knows much less (0.57)
 - ▶ After release (23 months): intervals include 0.5, but are too wide to prove anything
- ▣ Single days of Apple: AUC between 0.48 and 0.52 before release: no discrimination detected in this test, which does not prove that nothing is stored
 - ▶ Look-ahead is a danger for backtests at the horizon of well-known events, less for daily noise

Did the Memory Vanish? A Formal Test and Its Power

- $H_0 : \text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$, two independent samples; DeLong variance for each AUC (DeLong et al. (1988))
 - ▶ $z = (\widehat{\text{AUC}}_1 - \widehat{\text{AUC}}_2) / \sqrt{\widehat{V}_1 + \widehat{V}_2}$: 14B $z = 0.73$ ($p = 0.47$), 7B $z = 1.01$ ($p = 0.31$)
- Power: with 23 months after release, $\text{SE}(\text{AUC})$ under H_0 is 0.12; the smallest drop detectable with 80% power is 0.36
 - ▶ A fall from 0.69 to 0.5 is smaller than that: about 101 post-release months would be needed
- Not rejecting H_0 here says nothing: absence of evidence of memory after release is not evidence of its absence before

Remedies, Including Point-in-Time Models

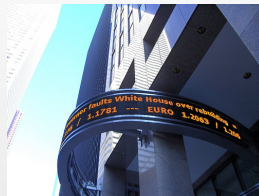
- Test only after the model's training cutoff: [Lopez-Lira & Tang \(2026\)](#) evaluate GPT-4 on post-cutoff headlines; clean, but the window starts late
- Mask company names and dates before scoring ([Glasserman & Lin \(2023\)](#)), then compare slopes with and without masking: a diagnostic, since products, events or quotes can still identify the episode
- Point-in-time models, trained only on text available up to date t ([Sarkar & Vafa \(2024\)](#))
 - ▶ [Drinkall et al. \(2024\)](#): a series of point-in-time LLMs, each uninformed about facts after its date; models and training data are public
 - ▶ [He et al. \(2025\)](#): ChronoBERT and ChronoGPT use only text available at each date; on next-day returns from news their Sharpe ratios are close to a much larger model: look-ahead there is modest
- Report the model version and the release date of its weights with every result

Section Summary

- LLMs trained on recent text carry knowledge of the past that a backtest must not use
- Qwen2.5-7B and 14B partly "know" the monthly direction of the S&P 500 before 2024, not single days
- Clean temporal tests use post-cutoff data or point-in-time models; anonymised text is an additional sensitivity check

Data: Dated Headlines and Prices

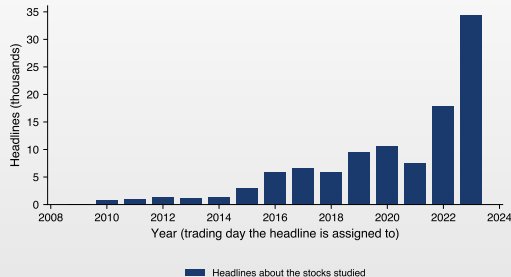
- ▣ FNSPID (Dong, Fan & Peng (2024)): 15.7 million news records about S&P 500 firms, 1999–2023, from four news websites
 - ▶ Licence CC BY-NC 4.0 (non-commercial use)
- ▣ Here: 16 large US stocks with daily prices in the course data (technology, banks, GameStop, Strategy)
 - ▶ 106 820 distinct headlines after removing repeats; 18 282 stock-days with news, 27.10.2009 to 18.12.2023



News ticker on the Reuters building, New York

Photo: bgilliard (2005); CC BY-SA 2.0; Wikimedia Commons

How Many Headlines, and When



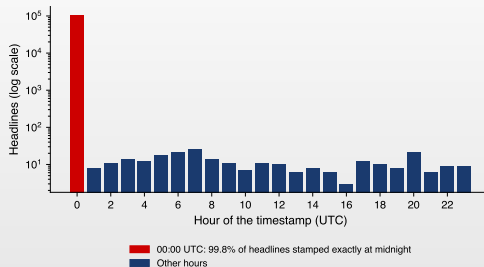
□ Headlines about the 16 stocks per year, by the trading day they are assigned to

 MFM_ch15_news_data

Interpretation: Coverage Is Uneven

- 34 380 headlines in 2023 alone, against 8 445 for all years before 2016
 - ▶ Results are dominated by recent years; early years have few news days per stock
- On a news day a stock has on average 5.8 headlines
 - ▶ Daily score: the average of the headline scores for that stock and day
- Survivorship bias: the 16 stocks belong to today's large firms, selected with hindsight

The Timestamp Problem



- Hour of the FNSPID timestamp in Coordinated Universal Time (UTC), log scale

 MFM_ch15_news_data

Interpretation: Dates, Not Times

- ▣ 99.8% of headlines are stamped exactly 00:00 UTC: the source gives the date, not the time of publication
 - ▶ A headline dated d may have appeared before the open, during the session or after the close of day d
- ▣ Alignment used here, with no look-ahead
 - ▶ Day d : the first trading day on or after the date of the headline
 - ▶ Returns on d and $d + 1$ may still contain the reaction; the first safe forecast is the return on $d + 2$ (close of $d + 1$ to close of $d + 2$)
- ▣ Excess return: stock return minus the return of SPY (an S&P 500 index fund) on the same day

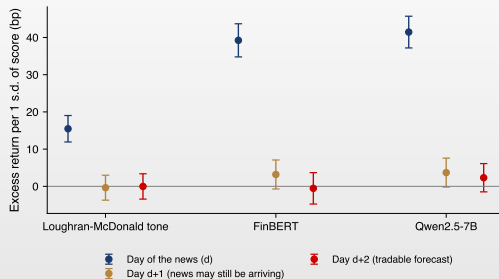
Three Scores for Every Headline

- LM tone; FinBERT net score $P(\text{pos}) - P(\text{neg})$; Qwen2.5-7B net score with prompt P1
 - ▶ LM finds a listed word in only 27% of headlines: the rest get tone 0
 - ▶ FinBERT labels: 19% negative, 62% neutral, 18% positive
- Correlations across headlines: FinBERT–Qwen 0.62, FinBERT–LM 0.41, Qwen–LM 0.36
 - ▶ The two language models agree much more with each other than with the dictionary
- Scores are standardised over the panel (mean 0, standard deviation 1) before regressions

Regression Design

- Pooled regression over stock-days with news: $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon_{i,d+h}$, $h = 0, 1, 2$
 - ▶ r^e : excess return (%); z : standardised daily score; b : return per one standard deviation of score
- Errors of stocks on the same day are correlated (common shocks): cluster standard errors by day (Petersen (2009))
 - ▶ $\hat{V} = (X^\top X)^{-1} \left(\sum_g X_g^\top \hat{\varepsilon}_g \hat{\varepsilon}_g^\top X_g \right) (X^\top X)^{-1}$, g : trading days
- $h = 0$ measures reaction, $h = 2$ measures predictability

Reaction versus Prediction



□ Slope b in basis points (bp) per one standard deviation of score, with 95% CI clustered by day

 MFM_ch15_news_signal

Interpretation: Headlines Describe Prices

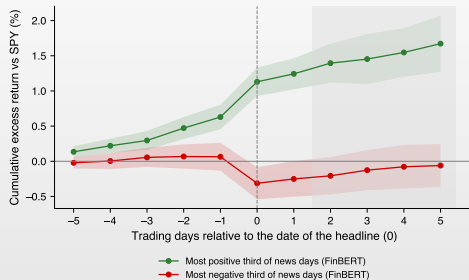
- ▣ Day d : FinBERT +39.2 bp ($t = +17.27$), Qwen +41.4 bp ($t = +19.07$), LM +15.5 bp ($t = +8.53$)
 - ▶ Many headlines report the price move itself ("shares jump", "stock falls"): causality runs both ways
- ▣ Day $d + 1$: FinBERT +3.2 bp ($t = +1.60$), Qwen +3.7 bp ($t = +1.86$)
 - ▶ Part of this is news published after the close of d , not a forecast
- ▣ Day $d + 2$: FinBERT -0.5 bp ($t = -0.25$), Qwen +2.3 bp ($t = +1.19$), LM -0.0 bp ($t = -0.01$)

Inference with 16 Stocks and Nine Regressions

Score, horizon	b (bp)	t by stock	t two-way	p wild bootstrap	p Holm
FinBERT, d	+39.2	+5.71	+5.67	<0.001	<0.001
Qwen, d	+41.4	+5.10	+5.07	<0.001	<0.001
FinBERT, $d + 2$	-0.5	-0.35	-0.30	0.739	1.00
Qwen, $d + 2$	+2.3	+2.41	+2.01	0.061	0.37
LM, $d + 2$	-0.0	-0.03	-0.02	0.979	1.00

- Firm effects persist over time, so errors also correlate within a stock: cluster by stock, or by day and stock ([Cameron, Gelbach & Miller \(2011\)](#)); [Driscoll & Kraay \(1998\)](#) add serial correlation
 - ▶ With 16 clusters the clustered SE tends to be too small: wild cluster bootstrap with the null imposed, Rademacher weights, 9,999 draws ([Cameron, Gelbach & Miller \(2008\)](#)); it assumes independent stocks, so read it next to the two-way t
- Qwen at $d + 2$: $t = +2.41$ by stock looks significant; bootstrap $p = 0.061$, and 0.37 after Holm over the 9 regressions (3 scores $\times$ 3 horizons)

Event Study around News Days



- Average cumulative excess return from day -5 to $+5$; news days split by tertiles of the daily FinBERT score; shaded: the tradable days $d + 2$ to $d + 5$

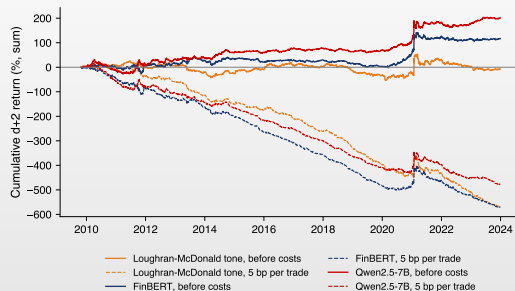
Interpretation: the Move Happens Before and On the News

- Most positive third: +0.63% over the 5 days before, +0.50% on day d ; most negative third: +0.06% and -0.38%
 - ▶ Pre-event moves: information leaks, or headlines written after the price has already moved
- The sentiment effect is the **difference** positive minus negative, pooled over stock-days with a CI clustered by date; the group figures above average each date first, so they need not differ by the same amount
 - ▶ Day d : +1.00% [+0.88; +1.13]; tradable days $d + 2$ to $d + 5$: +0.17% [-0.08; +0.42]
- Standardised test robust to event-induced variance (Boehmer, Musumeci & Poulsen (1991)) on $d + 2$ to $d + 5$: $z = +2.62$ (positive), +1.25 (negative)
 - ▶ Both thirds drift up; the selection of today's large firms may contribute, but these averages do not identify the cause; the test assumes independent events, so it overstates z

A Daily Sentiment Strategy

- At the close of $d + 1$: for every stock with news on d , buy if its score is positive and sell short if negative
 - ▶ Each position hedged with SPY; equal weights; close after one day and earn the excess return of $d + 2$
 - ▶ Days with no news: no position, return 0
- Costs: 5 bp per trade; four trades per day with positions (stock and SPY, in and out)
 - ▶ Break-even cost: mean gross return on days with positions divided by 4
- Inference: Newey–West t (Chapter 2) and block-bootstrap CI for the mean daily return

Cumulative Strategy Returns



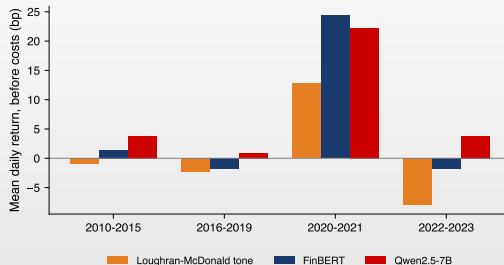
□ Sum of daily excess returns (%), before costs and with 5 bp per trade

 MFM_ch15_news_signal

Interpretation: a Small Signal, No Net Profit

- ▣ FinBERT: mean +3.27 bp per day ($t = +1.27$), CI 95% [-1.51; +8.93]: not different from zero
 - ▶ LM: -0.21 bp ($t = -0.07$)
- ▣ Qwen2.5-7B: +5.64 bp ($t = +2.26$, $p = 0.024$), CI 95% [+0.95; +10.61]; best of three scores: Holm $p = 0.071$, not significant
 - ▶ Break-even cost +1.5 bp per trade, well below 5 bp: Sharpe +0.63 before and -1.48 after costs
 - ▶ Qwen2.5 was trained after 2023: part of its edge may be memorised prices (see Look-Ahead Bias and Memorisation)
- ▣ Non-tradable day $d + 1$: FinBERT -0.22 bp ($t = -0.10$), Qwen +4.06 bp ($t = +1.36$)
 - ▶ No significant gain even there: almost all of the reaction happens on day d

Stability over Time



- Mean daily return of the strategy before costs, by subperiod

 MFM_ch15_news_signal

Interpretation: Weak Evidence, Stated Honestly

- ▣ FinBERT by subperiod: 2010–2015 +1.4 bp, 2016–2019 -1.8 bp, 2020–2021 +24.3 bp, 2022–2023 -1.8 bp
 - ▶ Only 2020–2021 is large (FinBERT $t = +1.89$, Qwen +22.2 bp, $t = +1.78$): not significant at 5%, two years of meme-stock trading
 - ▶ Four trades of 5 bp cost 20 bp per day: even that period would barely break even
- ▣ Why so little: 16 heavily followed stocks, date-only timestamps, daily frequency
 - ▶ Published evidence of predictability relies on precise timestamps and many small stocks
- ▣ A negative result is still a result: it tells you where the signal is not

What the Literature Finds

- ▣ **Lopez-Lira & Tang (2026)**: on headlines published after GPT-4's knowledge cutoff, GPT-4 captures the initial, non-tradable market reaction (abstract of [arXiv:2304.07619](https://arxiv.org/abs/2304.07619): portfolio-day hit rates of about 90%)
 - ▶ That initial reaction is not tradable; GPT-4 scores also predict the later drift, most for small stocks and negative news
 - ▶ Strategy returns decline over the sample; the authors read this as more efficient prices as LLM adoption rises, alongside changing market conditions (Section 8.2)
- ▣ **Bollen, Mao & Zeng (2011)**: collective Twitter mood predicted the Dow Jones in 2008; the result is sensitive to the sample and method

Social Media and Crowds: GameStop, 2021

- January 2021: retail investors coordinated on social media and pushed GameStop up by more than ten times in a few weeks
 - ▶ Attention, sentiment and trading rose together; this episode alone cannot separate prediction from causation: each needs its own test
- Lessons for text signals
 - ▶ Volume of attention can matter more than tone
 - ▶ Crowded signals revert and create risk for short sellers (Chapter 17: bubbles)



A GameStop store, Vallejo, California, February 2021

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia

[Commons](#)

Section Summary

- Align news and prices with care: date-only stamps push the first tradable return to $d + 2$
- Headline sentiment is strongly related to same-day returns, and barely to returns one can still trade
- Test signals with clustered errors (wild bootstrap with few clusters), Newey–West t , a multiplicity correction, costs and subperiods

Sentiment as a Generated Regressor

- Model: $r = a + \beta s^* + \varepsilon$ with true tone s^* ; we observe a model output $\hat{s} = s^* + u$
 - ▶ Classical error (u independent of s^* and ε): $\text{plim } \hat{\beta} = \beta \lambda$, $\lambda = \text{Var}(s^*) / [\text{Var}(s^*) + \text{Var}(u)] < 1$
 - ▶ The test of $\beta = 0$ stays valid; magnitudes are attenuated towards 0
- Standardised score z , non-differential error ($E[r \mid s^*, \hat{s}] = E[r \mid s^*]$): $\text{Cov}(r, z) = \beta \text{Cov}(s^*, z)$
 - ▶ Slope on $z = (\text{effect of one standard deviation of } s^*) \times \rho(s^*, \hat{s})$: divide by the correlation of truth and score
- Econometrics of AI-generated variables: Battaglia et al. (2024) (bias correction, joint estimation); Ludwig, Mullainathan & Rambachan (2024) (prediction versus estimation uses of LLMs)

Discrete Labels: Misclassification Is Not Classical Error

- Binary "negative news" dummy $s^* \in \{0, 1\}$, predicted label $\hat{s}$; $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
 - ▶ The error $u = \hat{s} - s^*$ is negatively correlated with s^* : the classical formula does not apply ([Aigner \(1973\)](#))
 - ▶ $\text{plim } \hat{b} = \beta \frac{\pi(1-\pi)(1-\alpha_0-\alpha_1)}{p(1-p)}$, $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$
- Twitter validation set, $\pi = 14.5\%$, from the confusion matrices
 - ▶ FinBERT: $\alpha_0 = 13.3\%$, $\alpha_1 = 23.9\%$, factor 0.45; Qwen2.5-7B: 0.59; LM ($\alpha_1 = 50.4\%$): 0.28
- As long as $\alpha_0 + \alpha_1 < 1$ the sign is kept, but a dummy slope can shrink to a third or less

How Much Attenuation in Our Slopes?

Score	$\rho(s^*, \hat{s})$, Twitter	b on day d (bp)	b/ρ (bp)	λ , one factor
LM	0.29	15	54	0.53
FinBERT	0.58	39	67	0.85
Qwen2.5-7B	0.71	41	58	0.78

- ρ : correlation of the net score with the human label $(-1, 0, 1)$; b/ρ : same-day effect of one standard deviation of true tone
 - ▶ After correction the three methods agree much more: the small LM slope is mostly measurement error
 - ▶ Assumes non-differential error and that the Twitter correlation carries over to FNSPID headlines
- λ : loading on the common factor of the three daily scores (next slide), a different benchmark from the human label

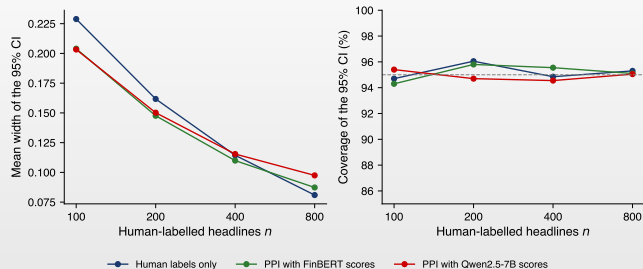
Two Noisy Measures: Factor Model and Instruments

- One factor: $z_j = \lambda_j f + u_j$, $\text{Var}(f) = 1$, u_j uncorrelated with f and with each other: $\rho_{jk} = \lambda_j \lambda_k$, so $\lambda_j^2 = \rho_{jk} \rho_{jl} / \rho_{kl}$
 - ▶ Daily FNSPID scores: $\lambda = 0.53$ (LM), 0.85 (FinBERT), 0.78 (Qwen)
- IV (instrumental variables): with $r = a + \beta f + \varepsilon$, instrument z_Q with z_F : $\hat{b}_{IV} = \widehat{\text{Cov}}(r, z_F) / \widehat{\text{Cov}}(z_Q, z_F) \rightarrow \beta / \lambda_Q$ if $\lambda_F \neq 0$, f is uncorrelated with ε , and u_F is uncorrelated with u_Q and ε
 - ▶ Day d , Qwen: 59 bp [53; 66] with FinBERT as instrument, 37 bp [28; 46] with LM (bootstrap over days)
- If both instruments were valid the two estimates would coincide: difference 22 bp, $z = 5.4$: rejected
 - ▶ The test rejects the joint restrictions, not a named instrument: FinBERT–Qwen errors correlated through overlapping training text are one plausible cause, an LM error related to returns another; the factor λ s are then suspect too

Valid Inference with Model Labels: Prediction-Powered Inference

- n texts labelled by humans (Y) and N unlabelled texts, independent random samples of one population; the model f is fixed (trained on other data)
 - ▶ PPI (Angelopoulos et al. (2023)): $\hat{\theta}_{\text{PPI}} = \frac{1}{N} \sum_{i \text{ unlab.}} f(\tilde{X}_i) - \frac{1}{n} \sum_{j \text{ lab.}} (f(X_j) - Y_j)$: prediction minus its estimated bias
 - ▶ $\widehat{\text{Var}} = \widehat{\text{Var}}(f)/N + \widehat{\text{Var}}(f - Y)/n$: an asymptotically valid CI under this sampling design, whatever the quality of f
- Narrower than labels alone when $\text{Var}(f)/N + \text{Var}(f - Y)/n < \text{Var}(Y)/n$, roughly $\text{Var}(f - Y) < \text{Var}(Y)$ when $N \gg n$
 - ▶ For a regression the rectifier applies to every moment condition that contains the predicted variable, also when it is a regressor (Egami et al. (2023))
- Egami et al. (2023): design-based supervised learning, a doubly robust version that allows labelling probabilities to depend on the text
- The few hundred labelled documents are not only training data: they are the validation sample that makes downstream inference valid

PPI on Twitter Headlines



- Target: mean human net tone $(-1, 0, 1)$; the 2388 validation headlines act as the population; 2,000 replications, n labelled and $2388 - n$ unlabelled headlines drawn independently, with replacement

 MFM_ch15_valid_inference

Interpretation: the Bias Correction Is the Main Gain

- True mean tone $\theta = 0.054$; plain model means are biased: FinBERT -0.014 (wrong sign), Qwen2.5-7B +0.101
 - ▶ No sample size fixes this: the model mean converges to the wrong number
- PPI keeps nominal coverage (94.3% at $n = 100$); CI width 0.204 against 0.229 with labels only
 - ▶ Variance reduction $1 - \text{Var}(f - Y)/\text{Var}(Y)$: 24% (FinBERT), 27% (Qwen): modest, because the models are noisy
 - ▶ Coverage stays near 95% for every n ; at $n = 800$ PPI is wider than labels alone (0.087 vs 0.081): with only 2388 – 800 unlabelled texts, $\text{Var}(f)/N$ is no longer negligible
- For the return regressions: label all headlines of a random sample of stock-days, so the true daily tone is known there, and rectify both OLS moment conditions (Egami et al. (2023))

Section Summary

- A sentiment score is an estimate: regressions on it are attenuated; dividing a standardised slope by $\rho(s^*, \hat{s})$ removes the attenuation
- Two model scores need not be independent measurements: the two instruments disagree, so at least one validity assumption fails; shared LLM errors are one candidate
- A small random human-labelled sample plus PPI gives valid intervals, whatever the model

Research Question: Does an LLM Read News as the Market Does?

- [Lopez-Lira & Tang \(2026\)](#) (*Journal of Financial Economics*): can a general-purpose LLM, with no explicit financial training, tell whether a headline is good or bad for the stock, and does its answer predict returns?
 - ▶ Initial reaction: the price move right after the news; not tradable, it measures whether the LLM understands the news
 - ▶ Drift: the return after the reaction, in the direction of the news; tradable, it measures underreaction of the market
- Link to this chapter: the same zero-shot scoring of headlines, but on news published after the model's training cutoff and with exact publication times
 - ▶ These are exactly the two weaknesses of our FNSPID test: date-only timestamps and look-ahead of Qwen2.5
- Why it matters: the LLM becomes a measuring instrument for how fast prices absorb news
 - ▶ Their model (Section 3): only LLMs above a quality threshold predict returns, and wide adoption should erode the predictability

Data and Method

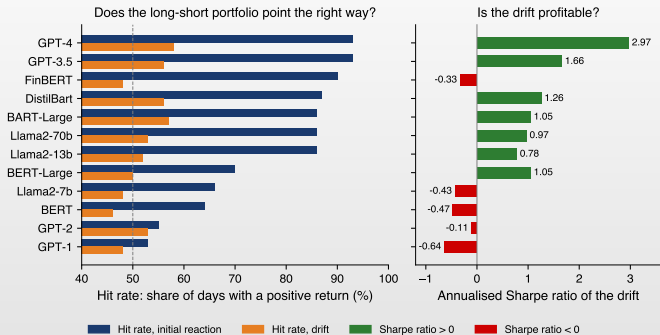
- 159,137 firm-headline-date observations, 4,123 US common stocks (NYSE, NASDAQ, AMEX), October 2021 – May 2024 (Section 4)
 - ▶ The sample starts after the September 2021 training cutoff of the scored model, gpt-4-0314 (Section 5)
 - ▶ Filters: news-vendor relevance score 100, no “stock-gain”/“stock-loss” headlines, only new events, near-duplicates of the same day removed
 - ▶ Overnight news (before 9 a.m. or after 4 p.m.): 82% of headlines; the rest are intraday
- Prompt (Section 5): the model acts as a financial expert and answers YES (good news), NO (bad news) or UNKNOWN on the short-term effect on the stock price; temperature 0
 - ▶ Score $s_{i,t}$: YES = 1, UNKNOWN = 0, NO = -1
- Portfolios (Table 1): each day, equal weights, long the stocks with a positive score and short those with a negative one, a leg only with at least two firms; for overnight news, initial reaction: previous close to open; drift: open to close
 - ▶ Hit rate: share of days with a positive portfolio return; Sharpe ratio $SR = \sqrt{252} \bar{r} / \hat{\sigma}$ (annualised)
- Regression, Eq. (1): $r_{i,t} = a_i + b_t + \gamma s_{i,t} + \varepsilon_{i,t}$, firm and date fixed effects, standard errors clustered by date and by firm (Table 2)
 - ▶ Twelve language models of increasing size scored with the same design, from GPT-1 to GPT-4 (Table 5)

Key Results

Model	Hit rate, reaction	Hit rate, drift	Drift, % per day	SR
GPT-4	0.93	0.58	0.34	2.97
GPT-3.5	0.93	0.56	0.29	1.66
FinBERT	0.90	0.48	−0.07	−0.33
GPT-1	0.53	0.48	−0.07	−0.64

- Overnight news, long-short portfolios (Table 5): GPT-4 gets the direction of the reaction right on 93% of days, like GPT-3.5, but earns the largest drift
 - ▶ Short leg 0.26% per day (SR 2.01), long leg 0.08% (SR 0.78) (Table 1): the market underreacts mainly to bad news
- Eq. (1), overnight news (Table 2): $\hat{\gamma} = 1.326$ ($t = 31.10$) for the reaction, 0.161 ($t = 6.80$) for the drift; adding the vendor's sentiment score leaves GPT-4 at 0.158 ($t = 5.35$) and the vendor at 0.010 ($t = 0.15$)
 - ▶ Small stocks: interaction coefficient 0.404 ($t = 4.75$): the drift is concentrated where arbitrage is costly
- Decay (Figure 8): drift SR 6.54 in 2021Q4, 3.68 in 2022, 2.33 in 2023, 1.22 in January–May 2024, as LLM use spread

Which Models Read the News Right, and Which Earn the Drift

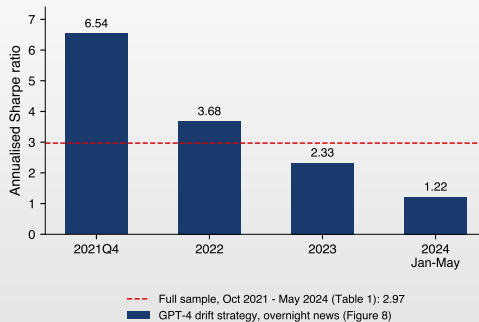


- Long-short portfolios on overnight news, twelve models with the same portfolio design (basic models and FinBERT scored with model-specific procedures, Appendix C), October 2021 – May 2024; source: [Lopez-Lira & Tang \(2026\)](#), Table 5

Interpretation: Understanding the News Is Necessary, Not Sufficient

- Threshold: GPT-1, GPT-2, BERT and Llama2-7b get the reaction right on only 53–66% of days, and their drift Sharpe ratios are negative (-0.64 to -0.11)
 - ▶ Seven models reach 86–93% on the reaction, but only GPT-4 pairs it with a drift Sharpe ratio near 3 (2.97, next GPT-3.5 at 1.66)
- FinBERT: 90% on the reaction, yet a drift Sharpe ratio of -0.33 ; it gives a direction for only 25% of overnight headlines (footnote 24)
 - ▶ Reading the obvious part of the news, already in the opening price, does not earn the drift
- Read the legs, not only the ratio: BERT-Large (Sharpe 1.05) holds on average 2 stocks in its short leg, GPT-3.5 holds 6, GPT-4 holds 20 (Table 5)
 - ▶ A high Sharpe ratio from a handful of short positions is fragile; compare with the macro-F1 by model size of Qwen2.5 in this chapter

The Drift Fades as LLMs Spread



- GPT-4 long-short drift strategy on overnight news, annualised Sharpe ratio by subperiod; source: [Lopez-Lira & Tang \(2026\)](#), Figure 8 and Section 8.2; full-sample value from Table 1

Interpretation: Adoption, or Something Else?

- The Sharpe ratio falls from 6.54 in 2021Q4 to 1.22 in January–May 2024, more than a fivefold drop; 2021Q4 and 2022 are above the full-sample 2.97
 - ▶ The full-sample estimate pools a strong start with a weak end (pooled mean over pooled volatility, not an average of the subperiod ratios); it is not a forecast of today's performance
- Their model (Section 3): as more investors process news with capable LLMs, underreaction and the drift shrink; the timing matches the spread of ChatGPT
 - ▶ The authors also name changing market conditions (Section 8.2); the subperiods are short (one quarter in 2021, five months in 2024), so the 95% intervals of Figure 8 are wide and the one for 2024 includes 0
- Intraday news, close-to-close drift: significant in 2022 and 2023, no longer in January–May 2024 (footnote 30)
 - ▶ A test that separates adoption from market conditions is still open: the extension on the next slide

Replication, Limitations and an Open Extension

- Closest course analogue: the Qwen2.5-7B strategy of this chapter on 16 FNSPID stocks earns +5.64 bp per day ($t = +2.26$), against 34 bp for GPT-4 (Table 1)
 - ▶ Replication (Seminar 15, C3): the published prompt, temperature 0, scores $\{-1, 0, 1\}$, the two-firm rule and Eq. (1) on FNSPID headlines and the adjusted closes of data/market
 - ▶ Data limits: date-only timestamps, so the first safe return is $d + 2$, not the next open; 16 large stocks; headlines that predate the model
- Limits stated by the authors: at 20 bp per round trip the strategy loses money (Figure 4), so it suits low-cost traders such as market makers (Section 1)
 - ▶ The decline over time may also reflect changing market conditions (Section 8.2); prompts and embedding models were not optimised, so the comparisons are lower bounds (Section 7.4)
- Open extension: on headlines published after the release of an open-weight model, does the drift survive, and does its Sharpe ratio keep falling as Proposition 2 predicts?

An Open Question: Can LLM-Generated Hypotheses Be Trusted?

- Ludwig & Mullainathan (2024): hypothesis generation as a statistical procedure: a model trained on data reveals a pattern, humans name it, fresh data test it
 - ▶ Their application: defendants' mugshots and judges' pretrial detention decisions
- Lopez-Lira, Tang & Zhu (2025): before its knowledge cutoff an LLM can recall outcomes, so a hypothesis it proposes about that period is not independent of the data that will test it
- Open: when an assistant proposes the signal, the prompt, the threshold and the holding period, what is the family of tests, and which inference stays valid? Agents that run the experiments themselves (Lu et al. (2024)) make the family larger and less visible
- Formally: $H_{0,j} : E[r_t^{(j)}] = 0$ for candidate strategies $j = 1, \dots, M$; control the FWER (Holm) or the FDR (Benjamini & Hochberg (1995))
 - ▶ M counts every variant the assistant proposed, including those discarded; this is not enough if the hypotheses were built on the test outcomes: the p -values then need untouched test data or selective (sequential) inference
 - ▶ Trading agents are covered in Practice and Conclusions; here the agent is a research assistant

The Discovery Loop, Step by Step (1/2)

- **Literature:** "Find studies since 2023 that predict stock returns with LLM sentiment and test after the model's knowledge cutoff: DOI or arXiv id, cutoff date, test window, headline estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search; [Lopez-Lira & Tang \(2026\)](#) is the reference design
- **Hypotheses:** "Propose mechanisms by which headline sentiment could predict returns after d+1 and, for each, the cross-sectional pattern that would falsify it."
 - ▶ Under-reaction predicts drift concentrated in small, less-covered stocks; memorisation predicts an edge that disappears after the release date
- **Data and code:** "From ch15_news_daily.csv and data/market, reproduce the daily Qwen strategy of this chapter (enter at the close of d+1, hedge with SPY) and its Newey-West t."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); the scores are saved, so no model needs to be re-run

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Before any new run, write the full grid of variants you would test and the multiple-testing correction; all of them will be reported."
 - ▶ Placebo: scores shuffled across stocks within the same day; subperiods; costs of 5 bp per trade
- **Critique**: "Write a referee report: every way this strategy could reflect look-ahead, memorisation, selection or timestamps, with the test for each."
 - ▶ Then check each proposed test yourself: a critique can also contain invented results
- **Report**: the whole grid, the variant chosen before the results, model versions and release dates

What the Researcher Must Check

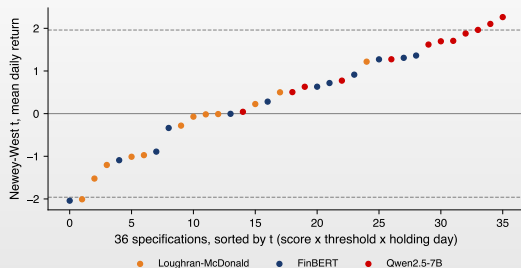
- **Memorisation:** the model knows what followed the headline
 - ▶ Test: news after the release of the weights; anonymised text as a diagnostic (Glasserman & Lin (2023)); the memory test of this chapter
- **Forking paths:** prompt, threshold, horizon tuned on the test period
 - ▶ Test: a pre-registered grid and a stated error rate: Holm (FWER) or BH (FDR, under independence or positive dependence); $|t| > 3$ (Harvey, Liu & Zhu (2016)) is a heuristic, not an error-rate guarantee
- **HARKing:** a plausible mechanism written after the result, with references that may not exist
 - ▶ Test: state the mechanism and its falsifier before the test; resolve every DOI
- **Timing:** date-only timestamps
 - ▶ Test: first tradable return on $d + 2$; the event study shows what is left on the tradable days
- **Agents that run code:** silent changes of data, windows or seeds
 - ▶ Test: read-only data, logged tool calls, the diff of every script before a result is accepted

Mini-Case: the Garden of Forking Paths

- A typical assistant proposal: "the Qwen signal works; refine it with a stronger threshold or a later entry day". We test the whole grid on the saved scores
- Grid: 3 scores (LM, FinBERT, Qwen) $\times$ 3 thresholds $|s_{i,d}| > b$, $b \in \{0, 0.25, 0.5\}$, on the raw score $\times$ 4 return days ($d + 2$ to $d + 5$, each a one-day position opened at the previous close) = 36 strategies
 - ▶ The base case replicates: 5.6 bp per day, $t = 2.26$
- $|t| > 1.96$ for 5 of 36 strategies, against 1.8 expected under the global null at 5%; 2 of them have the wrong sign
- Survivors: Holm (FWER 5%) 0, BH (FDR 10%) 0, $|t| > 3.0$; the Bonferroni threshold is $|t| > 3.20$
- Reading: the refinements add no evidence; the best variant is the one we started from, itself chosen among three scores, on a period the model may remember

 `ai_discovery_case.py`

The Specification Curve of the Grid



□ All 36 strategies of the mini-case sorted by t (Simonsohn, Simmons & Nelson (2020)); dashed: ± 1.96

 MFM_ch15_valid_inference

Interpretation: Is the Whole Curve Different from Noise?

- Joint null: no specification predicts returns; simulate it by flipping the signs of the positions at random in blocks of 10 consecutive signal days (dependent wild bootstrap, [Shao \(2010\)](#)), 2 000 times
 - ▶ The flips are shared by all 36 specifications: they keep the dependence across specifications and within blocks
 - ▶ Valid if, under the null, signed returns are symmetric around 0 and blocks are roughly independent
- Observed: 3 specifications with $t > 1.96$ (2 with $t < -1.96$), median $t = 0.50$
 - ▶ Under the null, 3 or more such specifications occur with probability 0.113; a median t this high, 0.207
- Reading: the curve is compatible with noise at 5%; reporting only the best point would hide this

Your Turn: a Project Seed

- Question: does LLM headline sentiment predict returns beyond $d + 1$ on news published after the model's release, under a pre-registered grid?
- Replicate first: the Qwen strategy of this chapter, +5.64 bp per day ($t = +2.26$); then the post-cutoff design of [Lopez-Lira & Tang \(2026\)](#)
- Deliverables
 - ▶ The pre-registered grid and correction; dated headlines after the release of an open model; the test on all variants
 - ▶ An LLM-proposed mechanism, stated before the test with its falsifier; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, memorised fact or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

AI Agents in Trading: Uses and Risks

- An agent is an LLM in a loop: read inputs, decide, call tools (data, code, news search, order entry through an API), observe, repeat
 - ▶ Used today for research summaries, coding and monitoring; fully autonomous trading is still rare and experimental
- Manipulation: a text read by the agent can carry instructions (prompt injection), e.g. a fake press release
- Systemic risk ([IMF \(2024\)](#)): many firms with similar models may trade alike, raising the speed and correlation of moves in stress
 - ▶ Controls: position limits, human approval of orders, logging of every step, tests after each model update

Pitfalls Checklist

- ▣ **Training overlap:** FinBERT scores 89.0% on PhraseBank, which largely overlaps its training data, 72.5% on new Twitter data
- ▣ **Look-ahead:** an LLM may remember the market after the news; test after its release date
- ▣ **Timing:** date-only timestamps; trade no earlier than the close after the news is surely public
- ▣ **Snooping:** prompts, thresholds and models chosen after seeing the test results
- ▣ **Costs and crowding:** small daily edges vanish after a few basis points of costs
- ▣ **Generated regressors:** scores are estimates; correct attenuation, validate on a random human-labelled sample (PPI)

When to Use What

Situation	Choice	Why
Fast, transparent baseline	LM dictionary	no training, auditable
Short English financial texts	FinBERT	accurate, milliseconds per text
No labels, other languages, long documents	open LLM (7B+)	zero-shot, flexible
A few hundred labels from your domain	embeddings + logistic regression	cheap and adapted to the domain

- ▣ Whatever the method: evaluate out of sample, with intervals and paired tests

Further Reading

- ▣ Tetlock (2007): media pessimism predicts market returns
- ▣ Loughran & McDonald (2011): a finance-specific sentiment dictionary
- ▣ Gentzkow, Kelly & Taddy (2019): text as data in economics
- ▣ Crypto applications: crypto-asset classification [Pele et al. \(2023\)](#), Dogecoin bubbles and sentiment [Pele \(2025\)](#), sentiment Quantlet [Pele \(2024\)](#)
- ▣ Further study: Quantinar: Natural Language Processing, Quantinar: DELTA, Quantinar: LDA Extensions; textbook [Franke, Härdle & Hafner \(2019\)](#) for the statistics behind the tests

Key Takeaways

- ▣ Generic dictionaries misread finance; the LM dictionary is a transparent baseline
- ▣ FinBERT and 7B–14B open LLMs show no significant accuracy difference on financial headlines (not proof of equality); a few hundred labels and a simple classifier compete with both
- ▣ LLMs remember market history: evaluate them only after their release date
- ▣ Headline sentiment explains same-day moves; with realistic timing and costs it does not beat the market in large stocks
- ▣ Scores are measurements with error: attenuation, possibly shared errors across models, few clusters and many tries all shape the inference

References (1/7)

- ▣ Aigner, D.J. (1973). Regression with a binary independent variable subject to errors of observation. *Journal of Econometrics*, 1(1), 49–59.
- ▣ Angelopoulos, A.N., Bates, S., Fannjiang, C., Jordan, M.I., Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671), 669–674.
- ▣ Antweiler, W., Frank, M.Z. (2004). Is all that talk just noise? The information content of internet stock message boards. *Journal of Finance*, 59(3), 1259–1294.
- ▣ Araci, D. (2019). FinBERT: financial sentiment analysis with pre-trained language models. arXiv:1908.10063.
- ▣ Baker, M., Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *Journal of Finance*, 61(4), 1645–1680.
- ▣ Battaglia, L., Christensen, T., Hansen, S., Sacher, S. (2024). Inference for regression with variables generated by AI or machine learning. arXiv:2402.15585.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Boehmer, E., Musumeci, J., Poulsen, A.B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.
- ▣ Bollen, J., Mao, H., Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1–8.

References (2/7)

- ▣ Brier, G.W. (1950). [Verification of forecasts expressed in terms of probability](#). *Monthly Weather Review*, 78(1), 1–3.
- ▣ Brown, T.B., Mann, B., Ryder, N., Subbiah, M., et al. (2020). [Language models are few-shot learners](#). *NeurIPS*; arXiv:2005.14165.
- ▣ Bybee, L., Kelly, B., Manela, A., Xiu, D. (2024). [Business news and business cycles](#). *Journal of Finance*, 79(5), 3105–3147.
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2008). [Bootstrap-based improvements for inference with clustered errors](#). *Review of Economics and Statistics*, 90(3), 414–427.
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2011). [Robust inference with multiway clustering](#). *Journal of Business & Economic Statistics*, 29(2), 238–249.
- ▣ Dawid, A.P., Skene, A.M. (1979). [Maximum likelihood estimation of observer error-rates using the EM algorithm](#). *Applied Statistics*, 28(1), 20–28.
- ▣ DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). [Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach](#). *Biometrics*, 44(3), 837–845.
- ▣ Devlin, J., Chang, M.-W., Lee, K., Toutanova, K. (2019). [BERT: pre-training of deep bidirectional Transformers for language understanding](#). *NAACL*; arXiv:1810.04805.
- ▣ Dong, Z., Fan, X., Peng, Z. (2024). [FNSPID: a comprehensive financial news dataset in time series](#). *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.

References (3/7)

- ▣ Drinkall, F., Rahimikia, E., Pierrehumbert, J.B., Zohren, S. (2024). [Time Machine GPT](#). arXiv:2404.18543.
- ▣ Driscoll, J.C., Kraay, A.C. (1998). [Consistent covariance matrix estimation with spatially dependent panel data](#). *Review of Economics and Statistics*, 80(4), 549–560.
- ▣ Egami, N., Hinck, M., Stewart, B.M., Wei, H. (2023). [Using imperfect surrogates for downstream inference: design-based supervised learning for social science applications of large language models](#). arXiv:2306.04746.
- ▣ Fleiss, J.L. (1971). [Measuring nominal scale agreement among many raters](#). *Psychological Bulletin*, 76(5), 378–382.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gentzkow, M., Kelly, B., Taddy, M. (2019). [Text as data](#). *Journal of Economic Literature*, 57(3), 535–574.
- ▣ Glasserman, P., Lin, C. (2023). [Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis](#). arXiv:2309.17322.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). [Empirical asset pricing via machine learning](#). *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q. (2017). [On calibration of modern neural networks](#). *ICML*; arXiv:1706.04599.

References (4/7)

- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- ▣ He, S., Lv, L., Manela, A., Wu, J. (2025). Chronologically consistent large language models. arXiv:2502.21206.
- ▣ Huang, A.H., Wang, H., Yang, Y. (2023). FinBERT: a large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2), 806–841.
- ▣ Hutto, C., Gilbert, E. (2014). VADER: a parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.
- ▣ IMF (2024). Advances in artificial intelligence: implications for capital market activities. *Global Financial Stability Report*, October 2024, Chapter 3.
- ▣ Jegadeesh, N., Wu, D. (2013). Word power: a new approach for content analysis. *Journal of Financial Economics*, 110(3), 712–729.
- ▣ Ke, Z.T., Kelly, B., Xiu, D. (2019). Predicting returns with text data. NBER Working Paper 26186.
- ▣ Kearney, C., Liu, S. (2014). Textual sentiment in finance: a survey of methods and models. *International Review of Financial Analysis*, 33, 171–185.
- ▣ Kim, A., Muhn, M., Nikolaev, V. (2023). Bloated disclosures: can ChatGPT help investors process information? arXiv:2306.10224.

References (5/7)

- ▣ Lopez-Lira, A., Tang, Y. (2026). [Can ChatGPT forecast stock price movements? Return predictability and large language models.](#) *Journal of Financial Economics*, 184, 104335. Earlier version: [arXiv:2304.07619](#).
- ▣ Lopez-Lira, A., Tang, Y., Zhu, M. (2025). [The memorization problem: can we trust LLMs' economic forecasts?](#) [arXiv:2504.14765](#).
- ▣ Loughran, T., McDonald, B. (2011). [When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks.](#) *Journal of Finance*, 66(1), 35–65.
- ▣ Loughran, T., McDonald, B. (2016). [Textual analysis in accounting and finance: a survey.](#) *Journal of Accounting Research*, 54(4), 1187–1230.
- ▣ Lu, C., Lu, C., Lange, R.T., Foerster, J., Clune, J., Ha, D. (2024). [The AI Scientist: towards fully automated open-ended scientific discovery.](#) [arXiv:2408.06292](#).
- ▣ Ludwig, J., Mullainathan, S. (2024). [Machine learning as a tool for hypothesis generation.](#) *Quarterly Journal of Economics*, 139(2), 751–827.
- ▣ Ludwig, J., Mullainathan, S., Rambachan, A. (2024). [Large language models: an applied econometric framework.](#) [arXiv:2412.07031](#).
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). [Good debt or bad debt: detecting semantic orientations in economic texts.](#) *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- ▣ Mikolov, T., Chen, K., Corrado, G., Dean, J. (2013). [Efficient estimation of word representations in vector space.](#) [arXiv:1301.3781](#).

References (6/7)

- ▣ Ouyang, L., Wu, J., Jiang, X., Almeida, D., et al. (2022). Training language models to follow instructions with human feedback. *NeurIPS*; arXiv:2203.02155.
- ▣ Pele, D.T. (2024). *Sentiment Analysis* (Quantlet, BitMood project). IDA Quantlets Repository, Bucharest University of Economic Studies.
- ▣ Pele, D.T. (2025). *Dogecoin and speculative bubbles: a study using LPPL and sentiment analysis*. Quantinar course, Bucharest University of Economic Studies.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets? The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Petersen, M.A. (2009). Estimating standard errors in finance panel data sets: comparing approaches. *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). *Qwen2.5 technical report*. arXiv:2412.15115.
- ▣ Reimers, N., Gurevych, I. (2019). Sentence-BERT: sentence embeddings using Siamese BERT-networks. *EMNLP*; arXiv:1908.10084.
- ▣ Sarkar, S.K., Vafa, K. (2024). Lookahead bias in pretrained language models. SSRN Working Paper 4754678.
- ▣ Shannon, C.E. (1951). Prediction and entropy of printed English. *Bell System Technical Journal*, 30(1), 50–64.

References (7/7)

- ▣ Shao, X. (2010). [The dependent wild bootstrap](#). *Journal of the American Statistical Association*, 105(489), 218–235.
- ▣ Shiller, R.J. (2017). [Narrative economics](#). *American Economic Review*, 107(4), 967–1004.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Taddy, M. (2013). [Multinomial inverse regression for text analysis](#). *Journal of the American Statistical Association*, 108(503), 755–770.
- ▣ Tetlock, P.C. (2007). [Giving content to investor sentiment: the role of media in the stock market](#). *Journal of Finance*, 62(3), 1139–1168.
- ▣ Tetlock, P.C., Saar-Tsechansky, M., Macskassy, S. (2008). [More than words: quantifying language to measure firms' fundamentals](#). *Journal of Finance*, 63(3), 1437–1467.
- ▣ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. (2017). [Attention is all you need](#). *NeurIPS*; arXiv:1706.03762.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). [MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers](#). *NeurIPS*; arXiv:2002.10957.
- ▣ Wu, S., Irsoy, O., Lu, S., Dabrowski, V., et al. (2023). [BloombergGPT: a large language model for finance](#). arXiv:2303.17564.